USN | | | | | | | | | | |

[23TCSE304]

# THIRD SEMESTER B.Tech EXAMINATION, SEPTEMBER-2024

## DATA BASE MANAGEMENT SYSTEMS

**Time: 3 Hours** **Maximum Marks: 100**

**Instructions:** Missing data may be suitably assumed.

**ANSWER ALL QUESTIONS**

| | | | | | |
|---|---|---|---|---|---|
| | | **PART-A** | **5x2=10** | | |
| **1.** | **a.** | What are the responsibilities and functions of a Database Administrator (DBA)? | L1 | CO1 | **02** |
| | **b.** | How does referential integrity help to maintain consistent relationships between tables in a relational database? | L3 | CO2 | **02** |
| | **c.** | What is the role of the COMMIT statement in SQL? and why is it important in database transactions? | L2 | CO3 | **02** |
| | **d.** | Consider the schema for Bank Database: Bank (BID, Bname, City, state) write an SQL query to retrieve BID and Bname located in Karnataka state. | L5 | CO4 | **02** |
| | **e.** | What are the potential anomalies that can arise in concurrency control? | L4 | CO5 | **02** |
| | | | | | |
| | | **PART-B** | **6 X 5=30** | | |
| **2.** | **a.** | What is a data model? Describe different types of data model used in DBMS | L2 | CO1 | **05** |
| | | **OR** | | | |
| | **b.** | Define database management system. List out the applications of DBMS. | L1 | CO1 | **05** |
| | | | | | |
| **3.** | **a.** | Illustrate select and project operations in relational algebra with an example. | L3 | CO2 | **05** |
| | | **OR** | | | |
| | **b.** | Explain in detail about Relational Model. | L2 | CO2 | **05** |
| | | | | | |
| **4.** | **a.** | Describe aggregate functions in SQL with examples. | L3 | CO3 | **05** |
| | | **OR** | | | |
| | **b.** | Write a short note on wildcard operators in SQL. | L3 | CO3 | **05** |
| | | | | | |

| | | | | | |
|---|---|---|---|---|---|
| **5.** | **a.** | Explain the GROUP BY statement and Having clause with an example. | L3 | CO4 | **05** |
| | | **OR** | | | |
| | **b.** | Write difference between a primary key and foreign key. | L4 | CO4 | **05** |
| | | | | | |
| **6.** | **a.** | Consider a relation R ( A , B , C , D , E , F, G ) with FDs { A → BC, BC → DE, D → F, CF → G} Find the closure of an attribute A. | L5 | CO5 | **05** |
| | | OR | | | |
| | **b.** | Explain with neat diagram the states for transaction execution. | L3 | CO5 | **05** |
| | | | | | |
| **7.** | **a.** | What are the data types in SQL? Explain with suitable example. | L3 | CO3 | **05** |
| | | **OR** | | | |
| | **b.** | Define the following terms:<br>**i.** physical schema<br>**ii.** logical schema<br>**iii.** Conceptual Schema | L2 | CO1 | **05** |
| | | | | | |
| | | **PART-C** | **6 X 10=60** | | |
| **8.** | **a.** | Construct an E-R diagram for a bank database by considering suitable entities and attributes. | L5 | CO1 | **10** |
| | | **OR** | | | |
| | **b.** | Draw and explain 3 tier architecture of DBMS. | L2 | CO1 | **10** |
| | | | | | |
| **9.** | **a.** | Consider a relation STUDENT shown below.<br>STUDENT (NAME, AGE, DEPARTMENT, CITY, MARKS)<br>Write Relational Algebra queries for the following statements.<br>**i.** Retrieve students' details those belongs to CSE department.<br>**ii.** Retrieve details of students whose department is CSE with age is more than 21.<br>**iii.** Retrieve name and department of the students whose marks is more than 90 and staying in Bangalore city.<br>**iv.** Retrieve name of the students studying in ECE department with age is more than 20.<br>**v.** Retrieve all the students details those who belongs to CSE with age is less than 20 or marks is more than 80. | L5 | CO2 | **10** |
| | | **OR** | | | |
| | **b.** | Illustrate relational algebra unary and binary operations with examples. | L3 | CO2 | **10** |

| | | | | | |
|---|---|---|---|---|---|
| | | | | | |
| **10.** | **a.** | Explain different types of SQL command Queries. | L3 | CO3 | **10** |
| | | **OR** | | | |
| | **b.** | Demonstrate SQL logical operators with examples. | L2 | CO3 | **10** |
| | | | | | |
| **11.** | **a.** | Define view/virtual tables in SQL with examples. | L3 | CO4 | **10** |
| | | **OR** | | | |
| | **b.** | Explain ALTER statement in SQL with examples. | L2 | CO4 | **10** |
| | | | | | |
| **12.** | **a.** | Explain the concept of normalization and discuss its benefits. | L4 | CO5 | **10** |
| | | **OR** | | | |
| | **b.** | Explain concurrency control with examples. | L2 | CO5 | **10** |
| | | | | | |
| **13.** | **a.** | Write short notes on the following key constraints in SQL.<br>**i.** NOT NULL<br>**ii.** UNIQUE<br>**iii.** PRIMARY KEY<br>**iv.** FOREGIGN KEY | L4 | CO4 | **10** |
| | | **OR** | | | |
| | **b.** | Explain in detail about Relational Model. | L3 | CO2 | **10** |